Multivariate Regression Model

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Multiple Regression Model

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Motivation

- **Example: Final exam score and missed classes**

  In the simple linear regression model, part of the error term would be student's ability.

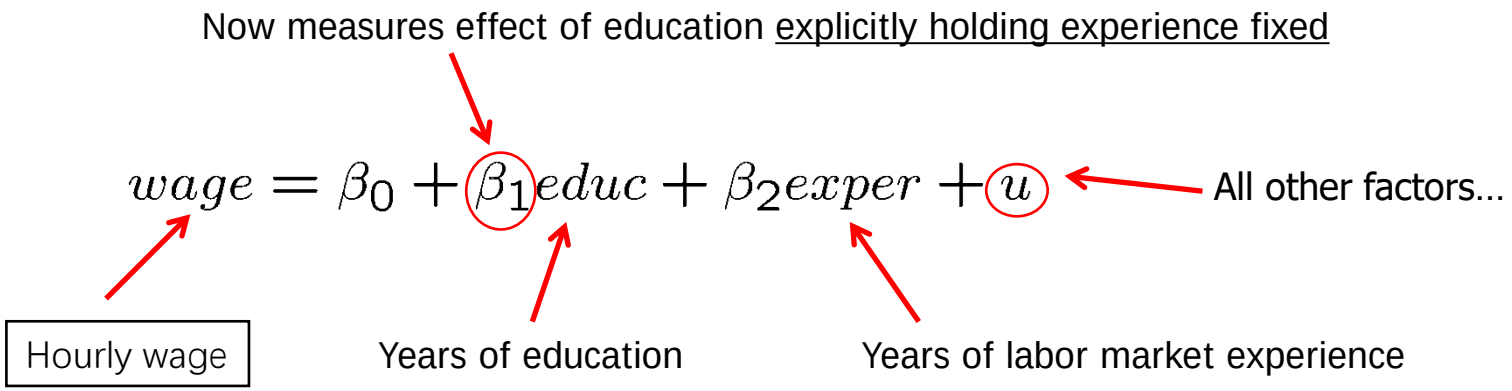
$$final = \beta_0 + \beta_1 missed + \beta_2 priGPA + u$$

where $priGPA$ is **grade point average** at the beginning of the term. We use it as a proxy for student's ability

$priGPA$ tends to be correlated with $missed$. In simple regression analysis, $priGPA$ would be included in the error term, which would be correlated with $missed$, causing the $\hat{\beta}_1$ to be biased.

Motivation

- Motivation for multiple regression
  - Incorporate more explanatory factors into the model
  - Explicitly hold fixed other factors that otherwise would be in u
  - Allow for more flexible functional forms
- Example: Wage equation



Motivation

- Example: Average test scores and per student spending

$$avgscore = \beta_0 + \beta_1 expend + \beta_2 avginc + u$$

Other factors

Average standardized test score of school

Per student spending at this school

Average family income of students at this school

  - Per student spending is likely to be correlated with average family income at a given high school because of school financing
  - Omitting average family income in regression would lead to biased estimate of the effect of spending on average test scores
  - In a simple regression model, effect of per student spending would partly include the effect of family income on test scores

Motivation

- Example: Family income and family consumption

$$cons = \beta_0 + \beta_1 inc + \beta_2 inc^2 + u$$

Family consumption

Family income

Family income squared

Other factors

  - Model has two explanatory variables: inome and income squared
  - Consumption is explained as a quadratic function of income
  - One has to be very careful when interpreting the coefficients:

By how much does consumption increase if income is increased by one unit?

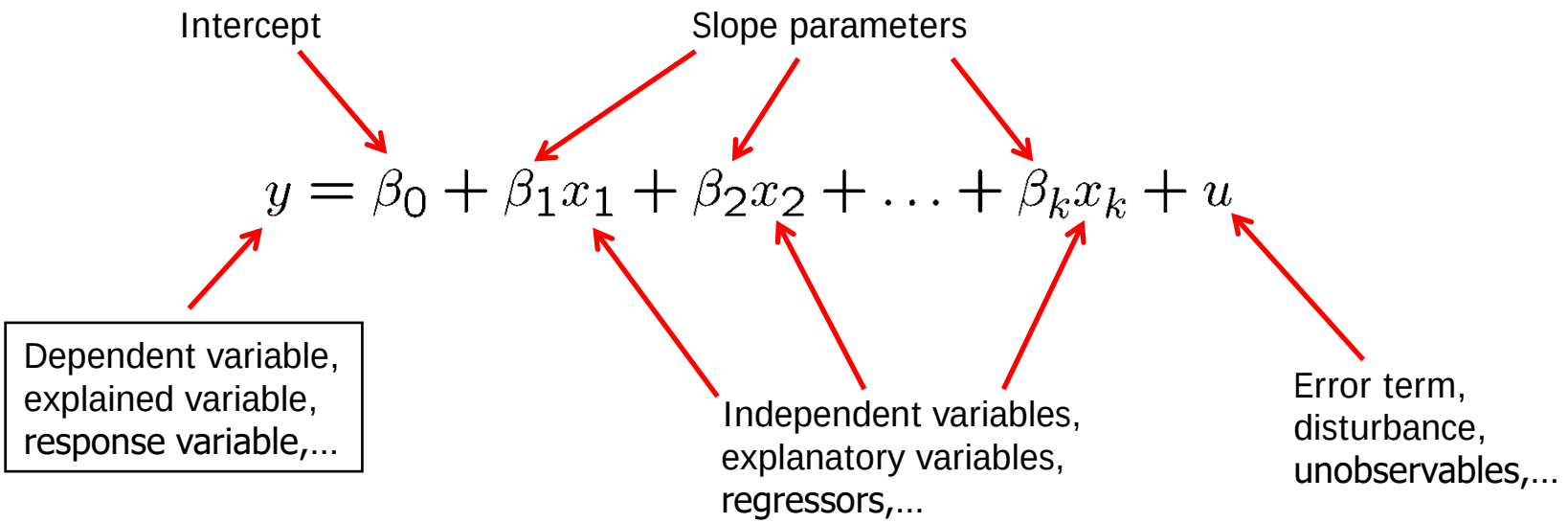
$$\frac{\Delta cons}{\Delta inc} \approx \beta_1 + 2\beta_2 inc$$

Depends on how much income is already there

Motivation

- Definition of the multiple linear regression model

"Explains variable y in terms of variables $x_1, x_2, \ldots, x_k$"



Motivation

- Example: CEO salary, sales, and CEO tenure

$$\log(salary) = \beta_0 + \beta_1 \log(sales) + \beta_2 ceoten + \beta_3 ceoten^2 + u$$

Log of CEO salary

Log sales

Quadratic function of CEO tenure with the firm

  - Model assumes a constant elasticity relationship between CEO salary and the sales of his or her firm
  - Model assumes a quadratic relationship between CEO salary and his or her tenure with the firm
- Meaning of "linear" regression
  - The model has to be linear in the parameters (not in the variables)

Motivation

- OLS Estimation of the multiple regression model
- Random sample

$$\{(x_{i1}, x_{i2}, \ldots, x_{ik}, y_i): \ i = 1, \ldots, n\}$$

- Regression residuals

$$\hat{u}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2} - \ldots - \hat{\beta}_k x_{ik}$$

- Minimize sum of squared residuals

$$\min \sum_{i=1}^{n} \hat{u}_i^2 \quad \rightarrow \quad \hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \ldots, \hat{\beta}_k$$

Minimization will be carried out by computer

Mechanics and interpretation of OLS

- **Mechanics and interpretation of OLS**

  Sample regression function with two independent variables

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

- **Obtaining the OLS estimates**

  Given random sample $\{(x_{i1}, x_{i2}, y_i) : i = 1, ..., n\}$. i is the observation number (as always) and the second subscript (1 and 2 in this case) are labels for particular variables. For example

$$\begin{aligned} x_{i1} &= educ_i,\ i = 1, ..., n \\ x_{i2} &= IQ_i,\ i = 1, ..., n \end{aligned}$$



Mechanics and interpretation of OLS

- **Deriving the OLS estimate**
- As in the simple regression case, the method of **ordinary least squares** chooses $\hat{\beta}_0, \hat{\beta}_1, \text{and } \hat{\beta}_2$ (so three unknowns) to minimize the sum of squared residuals,

$$\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2})^2$$

- The case with k independent variables is easy to state: choose the $k+1$ values $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \ldots, \hat{\beta}_k$ to minimize

$$\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \ldots - \hat{\beta}_k x_{ik})^2$$

Mechanics and interpretation of OLS

- Can be solved with multivariable calculus. The **OLS first order conditions** are the $k+1$ linear equations in the $k+1$ unknowns $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, ..., \hat{\beta}_k$

$$
\begin{aligned}
\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - ... - \hat{\beta}_k x_{ik}) &= 0 \\
\sum_{i=1}^{n} x_{i1}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - ... - \hat{\beta}_k x_{ik}) &= 0 \\
\sum_{i=1}^{n} x_{i2}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - ... - \hat{\beta}_k x_{ik}) &= 0 \\
&\vdots \\
\sum_{i=1}^{n} x_{ik}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - ... - \hat{\beta}_k x_{ik}) &= 0
\end{aligned}
$$

Mechanics and interpretation of OLS

- **Interpreting the OLS Regression Line**

  The slope coefficients now explicitly have **partial effect**, or **ceteris paribus**, interpretations.

  Consider

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

  Then

$$\Delta\hat{y} = \hat{\beta}_1 \Delta x_1 + \hat{\beta}_2 \Delta x_2$$

  so we can obtain the predicted change in y given the changes in x_1 and x_2
  For example:

$$\Delta\hat{y} = \hat{\beta}_1 \Delta x_1 \text{ , if } \Delta x_2 = 0$$

  similarly,

$$\Delta\hat{y} = \hat{\beta}_2 \Delta x_2 \text{ , if } \Delta x_1 = 0$$

Mechanics and interpretation of OLS

- **The model with *k* independent variables**

  We say that $\hat{\beta}_0, \hat{\beta}_1, ..., \hat{\beta}_k$ are the OLS estimates from the regression y on $x_1, x_2, ..., x_k$

- The OLS regression line is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + ... + \hat{\beta}_k x_k$$

  In terms of changes

$$\Delta\hat{y} = \Delta\hat{\beta}_0 + \Delta\hat{\beta}_1 x_1 + \Delta\hat{\beta}_2 x_2 + ... + \Delta\hat{\beta}_k x_k$$

- $\hat{\beta}_1$ measures the change in $\hat{y}$ due to a one-unit change in x_1, holding all other independent variables fixed. That is

$$\Delta\hat{y} = \hat{\beta}_1 \Delta x_1 \text{ , holding } x_2, x_3, ..., x_k \text{ fixed}$$

Mechanics and interpretation of OLS

- **Example: Final exam score and missed classes**

  Data: ATTNED.dta

  - Model 1:

$$\widehat{final} = 26.60 - .121\ missed$$
$$n = 680$$



  - Model 2:

$$\widehat{final} = 17.42 + .017\ missed + 3.24\ priGPA$$
$$n = 680$$

The coefficient on $priGPA$ means that one more point on prior GPA predicts a final exam score that is 3.24 points higher, holding $missed$ fixed.

Mechanics and interpretation of OLS

- **Example: Wage and education**

  Data: WAGE2.dta

  - Model 1:

$$\widehat{lwage} = 5.973 + .060\ educ$$
$$n = 935$$

  - Model 2:

$$\widehat{lwage} = 5.658 + .039\ educ + .006\ IQ$$
$$n = 935$$



Return to education holding IQ fixed, i.e., take two people, A and B, with the **same** IQ score. Suppose person B has one more year of schooling than person A. Then we predict B to have a wage that is 3.9% higher.

Mechanics and interpretation of OLS

- **What does it mean to "hold other factors fixed"**

  Multiple regression "partials out" the other independent variables when looking at the effect of, say, $educ$. It can show that $\hat{\beta}_1$ measures the effect of $educ$ on $lwage$ once the correlation between $educ$ and IQ has been partialled out.

IQ=100

	IQ	educ	wage
1.	100	13	1027
2.	100	12	1100
3.	100	12	650
4.	100	18	711
5.	100	12	400
6.	100	12	1202
7.	100	14	615
8.	100	12	769
9.	100	18	606
10.	100	12	1300

IQ=90

	IQ	educ	wage
1.	90	12	1000
2.	90	12	1151
3.	90	16	1346
4.	90	12	500
5.	90	13	500
6.	90	12	1063
7.	90	12	714
8.	90	12	879
9.	90	12	1366
10.	90	10	400

Mechanics and interpretation of OLS

- **A "partialling out" intepretation**

  Technically, if we focus on $\hat{\beta}_1$ in $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2}$, partialling out the effect of x_{i2} is the same as:

  1) Do a sample regression of x_{i1} on x_{i2} to obtain the residual $\hat{r}_{i1}$, where the effects of x_{i2} have been partialled out.

  2) Then regress y_i on $\hat{r}_{i1}$ to obtain $\hat{\beta}_1$.

  Therefore, $\hat{\beta}_1$ can be expressed as:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (\hat{r}_{i1} y_i)}{\sum_{i=1}^{n} (\hat{r}_{i1})^2}$$

- Example: Wage and education

Mechanics and interpretation of OLS

- "Partialling out" interpretation of multiple regression
- One can show that the estimated coefficient of an explanatory variable in a multiple regression can be obtained in two steps:
  - 1) Regress the explanatory variable on all other explanatory variables
  - 2) Regress y on the residuals from this regression
- Why does this procedure work?
  - The residuals from the first regression is the part of the explanatory variable that is uncorrelated with the other explanatory variables
  - The slope coefficient of the second regression therefore represents the isolated effect of the explanatory variable on the dep. variable

Mechanics and interpretation of OLS

- Interpretation of the multiple regression model

$$\beta_j = \frac{\Delta y}{\Delta x_j}$$

By how much does the dependent variable change if the j-th independent variable is increased by one unit, holding all other independent variables and the error term constant

  - The multiple linear regression model manages to hold the values of other explanatory variables fixed even if, in reality, they are correlated with the explanatory variable under consideration
  - "Ceteris paribus"-interpretation
  - It has still to be assumed that unobserved factors do not change if the explanatory variables are changed

Mechanics and interpretation of OLS

- Example: Determinants of college GPA

$$\widehat{colGPA} = 1.29 + .453\, hsGPA + .0094\, ACT$$

Grade point average at college

High school grade point average

Achievement test score

- Interpretation
  - Holding ACT fixed, another point on high school grade point average is associated with another .453 points college grade point average
  - Or: If we compare two students with the same ACT, but the hsGPA of student A is one point higher, we predict student A to have a colGPA that is .453 higher than that of student B
  - Holding high school grade point average fixed, another 10 points on ACT are associated with less than one point on college GPA

Mechanics and interpretation of OLS

- Properties of OLS on any sample of data
- Fitted values and residuals

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \ldots + \hat{\beta}_k x_{ik} \qquad \hat{u}_i = y_i - \hat{y}_i$$

Fitted or predicted values

Residuals

- Algebraic properties of OLS regression

$$\sum_{i=1}^{n} \hat{u}_i = 0 \qquad \sum_{i=1}^{n} x_{ij}\hat{u}_i = 0 \qquad \bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}_1 + \ldots + \hat{\beta}_k \bar{x}_k$$

Deviations from regression line sum up to zero

Covariance between deviations and regressors are zero

Sample averages of y and of the regressors lie on regression line

Mechanics and interpretation of OLS

- Goodness-of-Fit
- Decomposition of total variation

$$STT = SSE + SSR$$

- R-squared

$$R^2 \equiv SSE/SST = 1 - SSR/SST$$

Notice that R-squared can only increase if another explanatory variable is added to the regression

- Alternative expression for R-squared

$$R^2 = \frac{\left(\sum_{i=1}^{n}(y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})\right)^2}{\left(\sum_{i=1}^{n}(y_i - \bar{y})^2\right)\left(\sum_{i=1}^{n}(\hat{y}_i - \bar{\hat{y}})^2\right)}$$

R-squared is equal to the squared correlation coefficient between the actual and the predicted value of the dependent variable

Mechanics and interpretation of OLS

- Example: Explaining arrest records

Number of times arrested 1986

Proportion prior arrests that led to conviction

Months in prison 1986

Quarters employed 1986

$$\widehat{narr86} = .712 - .150\, pcnv - .034\, ptime86 - .104\, qemp86$$

$$n = 2,725, \quad R^2 = .0413$$

- Interpretation:
  - If the proportion prior arrests increases by 0.5, the predicted fall in arrests is 7.5 arrests per 100 men
  - If the months in prison increase from 0 to 12, the predicted fall in arrests is 0.408 arrests for a particular man
  - If the quarters employed increase by 1, the predicted fall in arrests is 10.4 arrests per 100 men

Mechanics and interpretation of OLS

- Example: Explaining arrest records (cont.)
  - An additional explanatory variable is added:

$$\widehat{narr86} = .707 - .151\, pcnv + .0074\, avgsen - .037\, ptime86 - .103\, qemp86$$

$$n = 2,725, \quad R^2 = .0422$$

Average sentence in prior convictions

R-squared increases only slightly

- Interpretation:
  - Average prior sentence increases number of arrests (?)
  - Limited additional explanatory power as R-squared increases by little
- General remark on R-squared
  - Even if R-squared is small (as in the given example), regression may still provide good estimates of ceteris paribus effects

Mechanics and interpretation of OLS

- **Example: Final exam score and missed classes**

$$\begin{aligned} \widehat{final} &= 26.60 - .121\ missed \\ n &= 680,\ R^2 = 0.0196 \end{aligned}$$

$$\begin{aligned} \widehat{final} &= 17.42 + .017\ missed + 3.24\ priGPA \\ n &= 680,\ R^2 = 0.1342 \end{aligned}$$

$missed$ and $priGPA$ together explain about 13.42% of the variation In the final exam score

- **Example: Wage and education**

$$\begin{aligned} \widehat{lwage} &= 5.973 + .060\ educ \\ n &= 935,\ R^2 = 0.0974 \end{aligned}$$

$$\begin{aligned} \widehat{lwage} &= 5.658 + .039\ educ + .006\ IQ \\ n &= 935,\ R^2 = 0.1297 \end{aligned}$$

The expected value of the OLS Estimators

- Standard assumptions for the multiple regression model
- Assumption MLR.1 (Linear in parameters)

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k + u$$

In the population, the relationship between y and the explanatory variables is linear

- Assumption MLR.2 (Random sampling)

$$\{(x_{i1}, x_{i2}, \ldots, x_{ik}, y_i) : \ i = 1, \ldots, n\}$$

The data is a random sample drawn from the population

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_k x_{ik} + u_i$$

Each data point therefore follows the population equation

The expected value of the OLS Estimators

- Standard assumptions for the multiple regression model (cont.)
- Assumption MLR.3 (No perfect collinearity)

> "In the sample (and therefore in the population), none of the independent variables is constant and there are no exact linear relationships among the independent variables."

- Remarks on MLR.3
  - The assumption only rules out perfect collinearity/correlation bet-ween explanatory variables; imperfect correlation is allowed
  - If an explanatory variable is a perfect linear combination of other explanatory variables it is superfluous and may be eliminated
  - Constant variables are also ruled out (collinear with intercept)

The expected value of the OLS Estimators

- Example for perfect collinearity: small sample

$$avgscore = \beta_0 + \beta_1 expend + \beta_2 avginc + u$$

In a small sample, avginc may accidentally be an exact multiple of expend; it will not be possible to disentangle their separate effects because there is exact covariation

- Example for perfect collinearity: relationships between regressors

$$voteA = \beta_0 + \beta_1 shareA + \beta_2 shareB + u$$

Either shareA or shareB will have to be dropped from the regression because there is an exact linear relationship between them: shareA + shareB = 1

The expected value of the OLS Estimators

- Standard assumptions for the multiple regression model (cont.)
- Assumption MLR.4 (Zero conditional mean)

$$E(u_i|x_{i1}, x_{i2}, \ldots, x_{ik}) = 0$$

The value of the explanatory variables must contain no information about the mean of the unobserved factors

  - In a multiple regression model, the zero conditional mean assumption is much more likely to hold because fewer things end up in the error

- Example: Average test scores

$$avgscore = \beta_0 + \beta_1 expend + \beta_2 avginc + u$$

If avginc was not included in the regression, it would end up in the error term; it would then be hard to defend that expend is uncorrelated with the error

The expected value of the OLS Estimators

- Discussion of the zero mean conditional assumption
  - Explanatory variables that are correlated with the error term are called endogenous; endogeneity is a violation of assumption MLR.4
  - Explanatory variables that are uncorrelated with the error term are called exogenous; MLR.4 holds if all explanat. var. are exogenous
  - Exogeneity is the key assumption for a causal interpretation of the regression, and for unbiasedness of the OLS estimators
- Theorem 3.1 (Unbiasedness of OLS)

$$MLR.1-MLR.4 \quad \Rightarrow \quad E(\hat{\beta}_j) = \beta_j, \quad j = 0, 1, \ldots, k$$

  - Unbiasedness is an average property in repeated samples; in a given sample, the estimates may still be far away from the true values

The expected value of the OLS Estimators

- **Unbiasedness of OLS (Chapter 3 Appendix)**

  In the model with *k* independent variables, we can write $\hat{\beta}_j$ as:

$$\hat{\beta}_j = \frac{\sum_{i=1}^{n}(\hat{r}_{ij}y_i)}{\sum_{i=1}^{n}(\hat{r}_{ij})^2} = \beta_j + \frac{\sum_{i=1}^{n}(\hat{r}_{ij}u_i)}{\sum_{i=1}^{n}(\hat{r}_{ij})^2}$$

  where $\hat{r}_{ij}$ is the residual from a regression of x_j on $x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_k$

$$E(\hat{\beta}_j) = \beta_j + E[\frac{\sum_{i=1}^{n}(\hat{r}_{ij}u_i)}{\sum_{i=1}^{n}(\hat{r}_{ij})^2}]$$

$$= \beta_j + \frac{\sum_{i=1}^{n}(\hat{r}_{ij}E(u_i))}{\sum_{i=1}^{n}(\hat{r}_{ij})^2} = \beta_j$$

The expected value of the OLS Estimators

- Including irrelevant variables in a regression model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$$

No problem because $E(\hat{\beta}_3) = \beta_3 = 0$.

= 0 in the population

However, including irrevelant variables may increase sampling variance.

- Omitting relevant variables: the simple case

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$

True model (contains x_1 and x_2)

$$\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1$$

Estimated model (x_2 is omitted)

The expected value of the OLS Estimators

- Omitted variable bias

$$x_2 = \delta_0 + \delta_1 x_1 + v$$

If x_1 and x_2 are correlated, assume a linear regression relationship between them

$$\Rightarrow \quad y = \beta_0 + \beta_1 x_1 + \beta_2(\delta_0 + \delta_1 x_1 + v) + u$$

$$= (\beta_0 + \beta_2\delta_0) + (\beta_1 + \beta_2\delta_1)x_1 + (\beta_2 v + u)$$

If y is only regressed on x_1 this will be the estimated intercept

If y is only regressed on x_1, this will be the estimated slope on x_1

error term

- Conclusion: All estimated coefficients will be biased

The expected value of the OLS Estimators

- **Omitting relevant variables: the simple case (cont.)**

Using

$$\tilde{\beta}_1 = \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)y_i}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2}$$

one can derive:

$$E(\tilde{\beta}_1) = \beta_1 + \beta_2 \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)x_{i2}}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2}$$

or

$$E(\tilde{\beta}_1) = \beta_1 + \beta_2 \tilde{\delta}_1$$

where $\tilde{\delta}_1$ is the slope coefficient in the simple regression: x_{i2} on x_{i1}, $i = 1, \ldots, n$

The expected value of the OLS Estimators

- Therefore, the bias in $\tilde{\beta}_1$ is:



$$Bias(\tilde{\beta}_1) = \beta_2 \tilde{\delta}_1$$

Called the **omitted variable bias**

- The simple regression estimator is unbiased in two cases:
  - $\beta_2 = 0$. But this means that x_2 does not appear in the population model.
  - $Corr(x_{i1}, x_{i2}) = 0$. Then the simple and multiple regression estimators are identical because $\tilde{\delta}_1 = 0$.
- Summary of bias in the simple regression estimator of $\tilde{\beta}_1$.

	$Corr(x_1, x_2) > 0$	$Corr(x_1, x_2) < 0$
$\beta_2 > 0$	Positive Bias	Negative Bias
$\beta_2 < 0$	Negative Bias	Positive Bias

The expected value of the OLS Estimators

- Example: Omitting ability in a wage equation

$$wage = \beta_0 + \beta_1 educ + \beta_2 abil + u$$

$$abil = \delta_0 + \delta_1 educ + v$$

Will both be positive

$$wage = (\beta_0 + \beta_2\delta_0) + (\beta_1 + \beta_2\delta_1) educ + (\beta_2 v + u)$$

The return to education β_1 will be overestimated because $\beta_2\delta_1 > 0$ It will look as if people with many years of education earn very high wages, but this is partly due to the fact that people with more education are also more able on average.

- When is there no omitted variable bias?
  - If the omitted variable is irrelevant or uncorrelated

The expected value of the OLS Estimators

- Omitted variable bias: more general cases

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$$ ← True model (contains x_1, x_2, and x_3)

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + w$$ ← Estimated model (x_3 is omitted)

  - No general statements possible about direction of bias
  - Analysis as in simple case if one regressor uncorrelated with others

- Example: Omitting ability in a wage equation

$$wage = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 abil + u$$

If exper is approximately uncorrelated with educ and abil, then the direction of the omitted variable bias can be as analyzed in the simple two variable case.